

Desafios Python

DESAFIO 1

```
# Estrutura base
emoji_data = {
    "nome": "Smile",
    "grade": [
        [(255,255,0), (255,255,0), (255,255,0), (255,255,0), (255,255,0)],
        [(255,255,0), (0,0,0), (255,255,0), (0,0,0), (255,255,0)],
        [(255,255,0), (255,255,0), (255,255,0), (255,255,0), (255,255,0)],
        [(255,255,0), (0,0,0), (0,0,0), (0,0,0), (255,255,0)],
        [(255,255,0), (255,255,0), (255,255,0), (255,255,0), (255,255,0)],
    ]
}

# Resultado: novo dicionário com sombra aplicada
emoji_sombra = {"nome": emoji_data["nome"], "grade": []}

# Nível 1: percorre o dicionário
for chave, valor in emoji_data.items():
    if chave == "grade":
        nova_grade = []

        # Nível 2: percorre cada linha da grade
        for linha in valor:
            nova_linha = []

            # Nível 3: percorre cada pixel da linha
            for pixel in linha:
                if pixel == (255, 255, 0): # pixel amarelo?
                    # Tuplas são imutáveis: criamos uma NOVA tupla
                    pixel_sombra = (pixel[0] // 2, pixel[1] // 2, pixel[2] // 2)
                    nova_linha.append(pixel_sombra)
                else:
                    nova_linha.append(pixel) # preto permanece

            nova_grade.append(nova_linha)

        emoji_sombra["grade"].append(nova_grade)

print(f"Emoji: {emoji_sombra['nome']}")
```

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for linha in emoji_sombra['grade'][0]:  
    print(linha)
```

DESAFIO 2

```
biblioteca_musical = {
    "Toques": [
        {"Alegre": ([440, 480], [520, 560])},
        {"Triste": ([200, 150], [100, 50])}
    ]
}

# Nível 1: percorre o dicionário principal com .items()
for secao, lista_toques in biblioteca_musical.items():
    print(f"Seção: {secao}")

# Nível 2: percorre cada dicionário de toque na lista
for toque in lista_toques:

    # Nível 3: percorre chave/valor de cada toque com .items()
    for nome, matriz in toque.items():
        print(f" {nome} original : {[list(linha) for linha in matriz]}")

    # Transposta: cada nova linha = uma coluna da original
    # A comprehension externa (for col) itera as colunas
    # A interna (for linha) coleta o valor de cada linha naquela coluna
    transposta = [
        [matriz[linha][col] for linha in range(len(matriz))]
        for col in range(len(matriz[0]))
    ]

    # .update() substitui o valor da chave no dicionário
    toque.update({nome: transposta})
    print(f" {nome} transposta: {transposta}")
```

DESAFIO 3

```
# Estrutura que mistura List, Tuple e Dict
playlist_imagens = {
    "nome": "Álbum Digital",
    "resolucao_padrao": (5, 5),      # Tupla: imutável
    "imagens": [                    # Lista: mutável
        {
            "titulo": "Smile",
            "grade": [
                [(255,255,0), (255,255,0), (255,255,0), (255,255,0), (255,255,0)],
                [(255,255,0), (0,0,0), (255,255,0), (0,0,0), (255,255,0)],
                [(255,255,0), (255,255,0), (255,255,0), (255,255,0), (255,255,0)],
                [(255,255,0), (0,0,0), (0,0,0), (0,0,0), (255,255,0)],
                [(255,255,0), (255,255,0), (255,255,0), (255,255,0), (255,255,0)],
            ]
        },
        {
            "titulo": "Coração",
            "grade": [
                [(0,0,0), (255,0,0), (0,0,0), (255,0,0), (0,0,0)],
                [(255,0,0), (255,0,0), (255,0,0), (255,0,0), (255,0,0)],
                [(255,0,0), (255,0,0), (255,0,0), (255,0,0), (255,0,0)],
                [(0,0,0), (255,0,0), (255,0,0), (255,0,0), (0,0,0)],
                [(0,0,0), (0,0,0), (255,0,0), (0,0,0), (0,0,0)],
            ]
        }
    ]
}

# Adiciona nova imagem com .insert() (método de lista)
nova_img = {
    "titulo": "Star",
    "grade": [
        [(0,0,0), (0,0,0), (255,255,0), (0,0,0), (0,0,0)],
        [(255,255,0), (255,255,0), (255,255,0), (255,255,0), (255,255,0)],
        [(0,0,0), (255,255,0), (255,255,0), (255,255,0), (0,0,0)],
        [(255,255,0), (0,0,0), (0,0,0), (0,0,0), (255,255,0)],
        [(255,255,0), (0,0,0), (0,0,0), (0,0,0), (255,255,0)],
    ]
}

playlist_imagens["imagens"].insert(1, nova_img) # .insert() – método de lista
# Remove a última imagem com .pop() (método de lista)
removida = playlist_imagens["imagens"].pop()
```

```
print(f"Removida: {removida['titulo']}")
```

```
# Nível 1: .keys() – itera chaves do dict principal
```